3D DETECTION · MULTI-VIEW · CAMERA-ONLY

Camera-Only 3D Detection

DETR3D → PETR → Sparse4D: three ways to lift multi-view 2D features into 3D space — explicit projection, precomputed position embedding, anchor-centric sparse propagation.

> **Central claim** — the fork in this lineage is the lift. DETR3D re-projects at every decoder layer, PETR folds 3D coordinates into a position embedding once, Sparse4D moves the lift onto anchors and keeps it alive across frames — O(T) stacking becomes O(1) recurrence.

47.9
DETR3D test NDS · V2-99

50.4
PETR test NDS · V2-99*

58.2
PETRv2 test NDS · V2-99

63.8
Sparse4D v2 test NDS

Problem — Where does "3D" come from in camera-only? › **Lift** — Projection matrices · 3D position embedding · 4D keypoints › **Time** — Single frame → 2 frames → stacked T → recurrent O(1)

1 · CITE
every number is a **paper table, abstract, or official README** cell — no third-party reposts, no own measurements on this topic.

2 · PIN
every recalled arXiv ID was re-checked on the abs page — **4 of 5 prior IDs were wrong** (e.g. PETR is 2203.05625, not 2203.12291).

3 · SPLIT
val vs test flagged on every row, with backbone and resolution. PETR's headline 50.4/44.1 is test · V2-99 · external data (Table 2).

4 · CROSS-CHECK
DETR3D test = **41.2 / 47.9** confirmed in its Table 2, README, and the comparison tables of PETR, PETRv2, and Sparse4D — four independent tables agree.

5 · SCOPE
survey only. Unverified recollections excluded: **"46.2/53.5"**, "GPU-center", **"15.9 FPS"**, Sparse4D "no NMS", "depth sort" — none appear in the sources.

Wang et al., CoRL 2021 — arXiv:2110.06922 · Liu et al., ECCV 2022 — arXiv:2203.05625
Liu et al., ICCV 2023 — arXiv:2206.01256 · Lin et al. (Horizon) — arXiv:2211.10581 · 2305.14018
Code: github.com/WangYueFt/detr3d · megvii-research/PETR · HorizonRobotics/Sparse4D

01 · FORMULATION

DETR3D: queries that go looking for pixels

A **sparse set of 3D object queries** is projected into each camera view via the transform matrix (Eq. 2), then features are bilinearly sampled; predictions use a **set-to-set loss**.

3D object queries › Project per camera › Sample 2D features

The re-projection happens **at every decoder layer** — lifting is paid repeatedly. Depth is never predicted, so the paper itself flags it: "depth estimation is still a core challenging" (§4.5).

NMS-free: "We do not use any post-processing such as NMS" (§4.1)

02 · DETR3D RESULTS

Cited numbers, conditions attached

CONFIG (PAPER)	SPLIT	MAP	NDS
R101+DCN, FCOS3D init †	val	34.6	42.5
R101+DCN, +CBGS #	val	34.9	43.4
V2-99, DD3D init #	**test**	**41.2**	**47.9**

FPS: not reported — only training cost (12 epochs, 8× RTX 3090, ~18 h). The famous "46.2 mAP / 53.5 NDS" pairing does not exist in any source we could check — **treated as invalid** and excluded.

repo: megvii/... deleted → WangYueFt/detr3d

03 · PETR

Kill the re-projection: 3D position embedding

Sample 64 points along the depth axis (**LID**, §4.2 — this depth mechanism belongs to PETR, **not** DETR3D). An MLP turns 3D coordinates into a **3D PE**, added to image features: "3D position-aware features".

The decoder is then a **standard DETR global-attention** pass over the flattened map — one shot, no per-layer projection. The lift cost moved from the decoder loop to a preprocessing step.

CONFIG	SPLIT	MAP	NDS
R101, 1408×512	val	35.7	42.1
R101	test	39.1	45.5
V2-99 * (external data)	**test**	**44.1**	**50.4**

10.7 FPS @ 1056×384 on V100 — vs BEVDet 4.2 FPS, but on a stronger 3090 (§4.4 flags the mismatch)

04 · PETRV2

The embedding learns to align time

The same 3D PE is extended for **temporal alignment of object positions** across frames, plus a feature-guided position encoder and **task-specific queries** (multi-task: BEV segmentation, 3D lane).

val R101 1600×640: 42.1 / 52.4 · test V2-99 49.0 / 58.2 · ms 50.8 / 59.1

2 frames is all it needs — yet it already beats Sparse4D's 9-frame val numbers on the test split. Venue verified in **CVF: ICCV 2023**, DOI 10.1109/iccv51070.2023.00302.

alignment moved from warping pixels to warping queries

05 · SPARSE4D

Anchors that look: 4D keypoints + depth reweight

Each 3D anchor carries **multiple 4D keypoints** (7 fixed + 6 learnable) projected to every view / scale / timestamp; features are fused hierarchically across views, scales, times, keypoints.

depth an **instance-level depth reweight module** eases the ill-posed 3D→2D projection (abstract). init **K-Means** on the training set: 900 anchors, 6 cascades, 640×1600.

CONFIG	SPLIT	MAP	NDS
R101, T=1	val	38.2	45.1
R101, T=9(–6)	val	44.5	54.7
Vov-99	**test**	**51.1**	**59.5**

06 · SPARSE4D V2

Stack → recurrence: O(T) to O(1)

In v1, temporal fusion **stacks T frames** — FPS falls 21.5 → 6.1 as T: 1→9 (v2 paper Table 1, R50·704×256·3090). v2 instead **propagates instances**: previous-frame outputs become current-frame anchors via ego motion, re-encoded each frame.

t−1 instances › Ego-motion re-projection › Anchor re-encode

EDA FPS 13.7 → **20.3** · inference memory **–53%** · training memory 6328 → 3100 MB (Table 2, 3090·batch 1)

The venue for v1/v2 is **unverified** — cite as arXiv preprint (Horizon Robotics); the integrated line appears in TPAMI 2026 per the official README.

07 · EVOLUTION TABLE

Same figure of merit, conditions attached

MODEL	TEST MAP	TEST NDS	KEY MOTION
DETR3D # (V2-99)	41.2	47.9	re-project each layer
PETR * (V2-99)	44.1	50.4	PE once, one-shot decode
PETRv2 (V2-99)	49.0	58.2	PE aligns time · 2 frames
Sparse4D (Vov-99)	51.1	59.5	anchor keypoints · stack T
Sparse4D v2 (Vov-99)	**55.7**	**63.8**	**propagate · O(1)**

All five are **external-data / large-backbone** configurations — the trend is real, direct cross-model speed claims are not.

test NDS: 47.9 → 50.4 → 58.2 → 59.5 → 63.8

08 · SPEED DISCIPLINE

FPS only travels with its GPU

Every FPS cell in this poster carries its conditions — none are directly comparable across papers:

CELL	CONDITIONS
PETR 10.7 FPS	V100 · 1056×384 (BEVDet 4.2 on 3090)
v1 21.5 → 6.1 (T=1→9)	3090 · 704×256 · R50 (Table 1)
v2 20.3 (any T)	3090 · 704×256 · R50 — same rig as v1
v1 2.9 / v2 8.4 (val)	README rig: 8× RTX 3090, own res/backbones

DETR3D and PETRv2 report **no FPS at all**. The only same-rig comparison is v1 vs v2: the O(1) design is what buys the speed, nothing else.

09 · OPEN LIMITS

What this poster does not claim

- **Validated excludes** — "46.2/53.5" (DETR3D), "GPU-center", "15.9 FPS", Sparse4D "no NMS" wording, "depth sort": none exist in the sources; kept out on purpose.
- **test rows use external data / V2-99** — trend comparison, not a controlled ablation chain.
- **val rows differ in backbone and resolution** — cited as published, not normalized by us.
- **Sparse4D v1/v2 venue unverified** — official indexes show preprint only; we say so instead of guessing.
- **LiDAR-free only** — BEVFormer and stream successors (StreamPETR, SparseBEV) are out of scope, listed as next.

Next: BEVFormer · StreamPETR · SparseBEV · day09_3d_fsd/01_초안.md